Multicloud Strategy & Architecture (1500+ words)

Multicloud Definition

Using multiple cloud providers to optimize cost, performance, risk, and workload specialization. Reduces vendor lock-in while enabling best-of-breed services.

Business Drivers

Cost optimization through competitive pricing. Performance through geographic distribution. Risk mitigation through vendor diversification. Workload specialization (e.g., ML on AWS, databases on Azure).

ABC Architecture

Primary: Azure (production, DR, CI/CD). Secondary: AWS evaluation for ML/analytics workloads. Hybrid: On-premises for regulated data.

Cloud Portability

Kubernetes standardizes container infrastructure across clouds. Service mesh provides vendor-agnostic service communication. Cloud-agnostic databases (PostgreSQL, MySQL) support portability.

Multicloud Governance

Unified cost management spanning clouds. Consistent tagging standards across providers. Federated authentication and identity. Compliance and security posture management.

Disaster Recovery

RPO (Recovery Point Objective): 1 hour max data loss. RTO (Recovery Time Objective): 4 hours max restore time. Warm standby in secondary cloud. Database replication with 1-hour lag.

Networking

VNets with region peering. ExpressRoute for dedicated Azure connectivity. Cross-cloud VPN. Data transfer cost: $0.02/GB.

Cost Considerations

Redundancy adds 20-40% infrastructure cost. Data transfer between clouds expensive. Negotiate volume commitments with both providers.

Roadmap

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Phase 1 (Current): Azure single-cloud focus. Phase 2 (Q2-Q3): AWS evaluation and pilot. Phase 3 (Q4+): Production multicloud with ML on AWS.